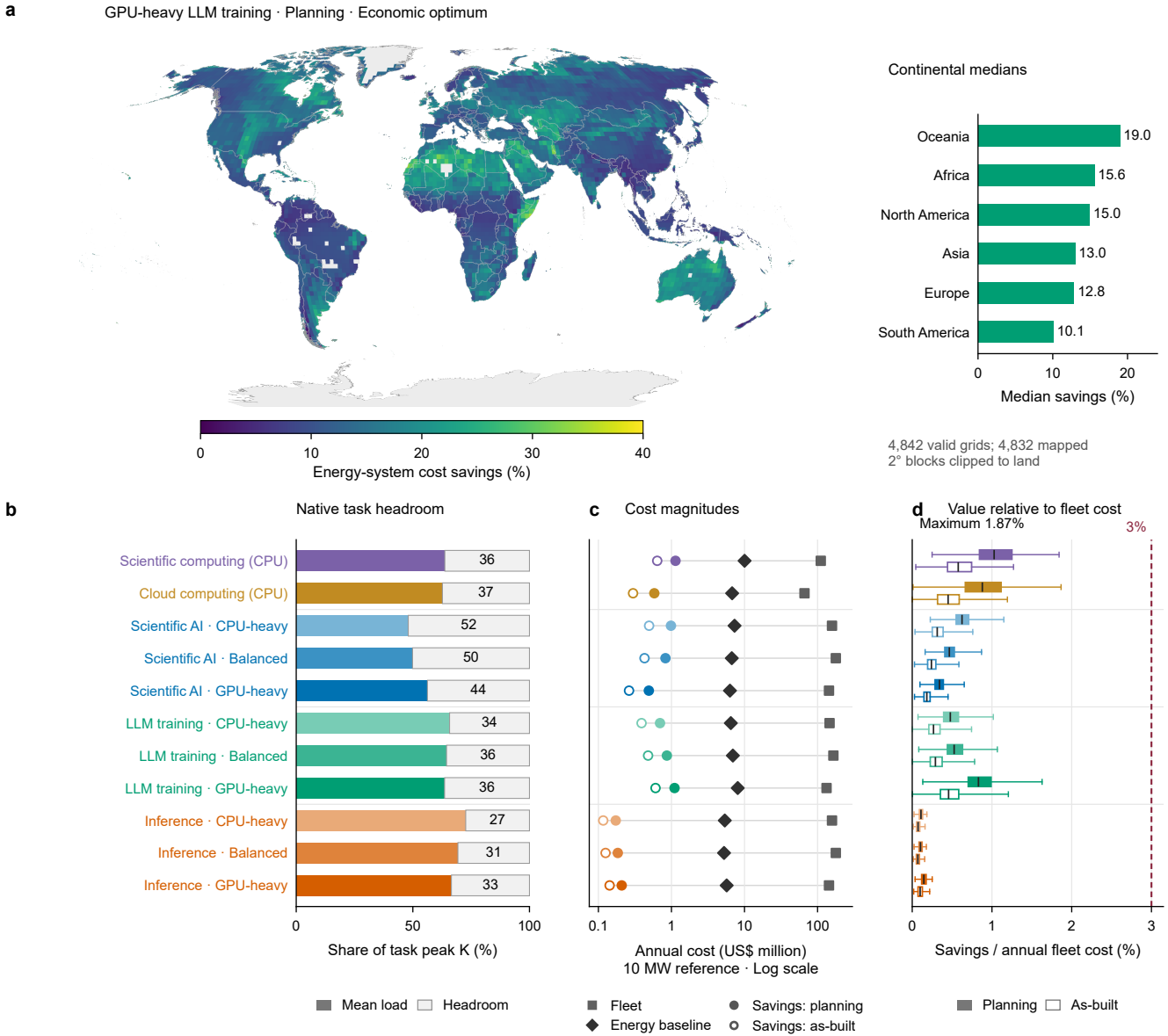

a
GPU-heavy LLM training · Planning · Economic optimum
0
10
20
30
40
Energy-system cost savings (%)
Continental medians
Oceania
19.0
Africa
15.6
North America
15.0
Asia
13.0
Europe
12.8
South America
10.1
0
10
20
Median savings (%)
4,842 valid grids; 4,832 mapped
2° blocks clipped to land
b
Native task headroom
Scientific computing (CPU)
36
Cloud computing (CPU)
37
Scientific AI · CPU-heavy
52
Scientific AI · Balanced
50
Scientific AI · GPU-heavy
44
LLM training · CPU-heavy
34
LLM training · Balanced
36
LLM training · GPU-heavy
36
Inference · CPU-heavy
27
Inference · Balanced
31
Inference · GPU-heavy
33
0
50
100
Share of task peak K (%)
Mean load
Headroom
c
Cost magnitudes
0.1
1
10
100
Annual cost (US$ million)
10 MW reference · Log scale
Fleet
Energy baseline
Savings: planning
Savings: as-built
d
Value relative to fleet cost
Maximum 1.87%
3%
0
1
2
3
Savings / annual fleet cost (%)
Planning
As-built